

CSCI 654 Projects

Timeline etc.

- 10/23/25 – Finalize project selections and groups
 - Please note that Project 5 must be completed solo.
- 11/20/25 Project reviews
- 12/05/25 - Final project and report due
- 12/11/25 – Project Demos

Projects 1 – Dynamic Scheduling

- Using AI to improve dynamic scheduling in Parallel Computing
- Most scheduling schemes follow this principle
 - With GPUs
 - Compute- intensive parts of applications are fully offloaded to a GPU
 - Workload is statically partitioned between CPUs and GPUs, with each partition running independently
 - Allow the GPU to exploit data parallelism
 - With Clusters/networks
 - Statically allocate parallel process components to parallel resources
 - No changes to address dynamics
- Project outline
 - Utilize Problem and system profiles
 - Implement offline/online machine learning algorithms to learn dynamically changing profiles
 - Dynamically allocate resources
 - Measure performance time (plus possibly energy consumption); e.g., IoT

Project 2, 3 –ANN parallel methods

- ANNs
 - Data
 - Similar to matrix-vector operations
 - Layer-wise parallelism
 - Allocate different layers to computing resources
 - Pipelining
 - Groups of layers
- CNNs
 - Across filters
- Project outline
 - Choose an application problem (discuss)
 - Discuss details
 - Measure performance time (plus possibly energy consumption); compare (time, accuracy) with serial and different methods for different sizes
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Project 4 – Federated learning and privacy

- Deep learning has two major challenges
 - Computational cost
 - Privacy
- Federated learning
 - Local training with local data
 - Encrypted outcome to master/cloud
 - Collective training using encrypted data in the cloud

Project 5 –Constructing Optimal Binary Search Trees

- **Project Problem**
- OBST is a binary search tree that provides the smallest expected search time for a given sequence of access probabilities. Please refer to notes on OBST and parallelization on myCourses under ‘contents’ for more details.
- Write a parallel program to construct OBSTs for 1) 500, 2) 1000 and 3) 1500 keys. You may use a suitable random generator to generate access probabilities or adopt access probabilities based on previous study by other/s. Improve your parallel program to when new keys are added or removed from the given set.
- OBST should be constructed using a parallel DP program. Implement your parallel program on 4 and 8 Cores, using shared memory or message passing programming paradigm. Compare with time taken for construction of OBST using a serial program. Present results in a table. Each data point should be an average of at least 19 runs.

Notes on OBSTs in slides 8 through 20.

This project must be completed solo. Not for groups.

Project 6 - LLMs and Parallelism

- Details to be inserted

Project 7 Parallelizing Blockchain computations

- Details to be inserted

Parallel systems in CS

Machine	OS	CPU	Memory	Video	GPU	notes
tardis	Debian 12	1 x 4-core Xeon CPU E3-1220 v5 @ 3.00GHz	64GB	ASPEED Graphics Family	none	amd64;openmpi teaching -- Head Node
kraken	Debian 12	2 x 22-core Xeon CPU E5-2696 v4 @ 2.20GHz	128GB	ASPEED Graphics Family	x4 Tesla K40c	amd64;openmpi teaching; GPU